

Research on Machine Translation for Titles of Scientific Papers with Contextual Information

Abstract

A paper's title is its summary of the central content and core viewpoints with concise and professional language to express its main idea and help readers quickly clarify its topic. Machine translation of titles is not only conducive to the rapid acquisition of researchers' desired cross language literature and improvement of work efficiency, It also facilitates knowledge service providers to provide cross language knowledge services and promotes global knowledge sharing.

This paper focuses on the machine translation of scientific papers' titles from Russian to Chinese. Titles are short texts containing specialized terminology, and their semantic expression is often limited due to the lack of contextual information. Existing research on title translation has attempted to address this issue by incorporating external knowledge or corpora, but it is a challenge to ensure that the added information is relevant and specialized to the titles. Moreover, the availability of Russian-to-Chinese parallel corpora is relatively limited compared to Russian-to-English and English-to-Chinese parallel corpora. Therefore, English is commonly used as a pivot language to enhance the machine translation performance between Russian and Chinese. However, existing pivot-based transfer learning methods using direct transfer in the pretraining-finetuning mode suffer from input and output word mismatches.

To address these challenges, this paper proposes a method that integrates context into the machine translation of scientific papers' titles from Russian to Chinese. The study consists of three parts: (1) **Introducing a context acquisition method based on large language model**. Professional vocabulary hidden in scientific papers' abstracts is highly related to the terminology in titles, so Abstracts can provide the most specialized context supplement to titles. This paper employed multiple strategies to obtain abstracts. When existing paper database query mechanisms and web retrieval mechanisms fail to obtain the real abstract for the title, an abstract generation model based on large language models is implemented by using various prompts for large language model pretraining-finetuning to generate professionally similar abstracts to the title. Experimental results show that the abstract information obtained

from the large language model achieves a Rouge-1 score of approximately 90, effectively meeting the demand for professional context supplementation for titles. (2) **Proposing a Russian-to-Chinese neural machine translation model based on pivot word embeddings.** Initially, Russian-to-English and English-to-Chinese machine translation models are pretrained on massive data from Russian-English and English-Chinese parallel corpora, respectively. Then, the encoder parameters of the Russian-to-English model and the decoder parameters of the English-to-Chinese model are transferred and fine-tuned on a small-scale Russian-to-Chinese parallel corpus. To address word mismatch issues during the transfer, a pivot word embedding generator is introduced into the transfer learning network. Experimental validation shows that the model's Bleu4 score outperforms the strong baseline system by 3.14 points, displaying strong openness and robustness, with a significant reduction in out-of-vocabulary words. (3) **Constructing a Russian-to-Chinese machine translation model for scientific papers' titles that incorporates contextual information.** Utilizing the two aforementioned methods to obtain initial translations of titles and contextual information translations, this paper gave a multi-encoder integration translation model. External and internal integration strategies are adopted to integrate context representations into the title translation model, generating more accurate final translations of titles. Experimental results shows that the model's Bleu4 score outperforms the strong baseline system by 1.22 points. The combination translation model effectively comprehends the semantic information for titles, significantly enhancing the quality of title translations.

In conclusion, the combination of context into Russian-to-Chinese machine translation of scientific papers titles effectively supplements professional context information of titles. English as a pivot language enables to obtain high-quality initial translations of titles and context translations, thereby enhancing the Russian-to-Chinese machine translation of scientific papers' titles.

There are 23 pictures and 14 tables in this study, with 81 references and 65 references in English.

Keywords: Low-resource machine translation; Transfer learning; Word embedding generator; Large language model; Context integration